

Table 6: List of all tested rules of Google’s Car App Quality Guidelines in AUTOGUARD. The specific rules mentioned in this paper have been bolded.

Categories	Criteria	App	ID & Description
General Functionality	Permitted Categories	Media,Navigation,POI, IOT,Messaging,Video, Games,Browsers,Weather	PC-1: The app must work as expected and not include features outside the app types.
	Expected Performance	Media,Navigation,POI, IOT,Video,Games, Browsers,Weather	EP-2: When the app is relaunched from the home screen, the app must restore the app state as closely as possible to the previous state.
	Irrelevant notifications	Media,Messaging,POI, Navigation,IOT,Weather	IN-1: The app must display notifications only when relevant to the driver’s needs.
		Video,Games,Browsers	IN-2: The app must not post any heads up notifications.
	Visual or Text Ads	Media,Messaging,POI, Navigation,IOT,Weather	AD-1: The app must not display text-based advertising other than the advertiser’s name or the product name.
	Notification Ads	Media,Messaging,POI, Navigation,IOT,Weather	NA-1: The app must not present advertisements through notifications.
	Messaging Functionality	Messaging	MF-1: The app must receive incoming messages.
			MF-2: Messages must be properly grouped and displayed in the correct order.
	Navigation Functionality	Navigation	MF-5: The app must implement a peer-to-peer messaging service and not notification services, such as those for weather, stocks, and sport scores.
			NF-2: The app draws only map content on the surface of the navigation templates. Text-based turn-by-turn directions, lane guidance, and estimated arrival time must be displayed on the relevant components of the navigation template. Additional information relevant to the drive, speed limit, road obstructions, etc., can be drawn on the safe area of the map.
	POI Functionality	POI	PF-1: The app must provide meaningful functionality relevant to driving.
	Weather Functionality	Weather	WE-1: App must include weather related content, which must be relevant to the user’s current location or a user specified location.
			WE-2: Weather information on map tiles must be readable and may not include complex legends. Apps may include a maximum of three legends. Apps with multiple legends may have a maximum of three colors, whereas single legend apps may have more than three colors.
			WE-5: Weather apps must not show more than five unique weather map annotations in a given view (for example: Temperature markers, wind speed markers, humidity, radar overlay, lightning indicators, road conditions all in the same view).
	Media Controls	Video	MC-1: The app integrates with media session. Depending on the content, the app must support either the play/pause or stop playback commands. Additionally, the app must provide title and thumbnail metadata for every media item.
Specific Scenarios	Visual Information on Phone	Media,Navigation,POI, IOT,Weather	VI-1: If the user must go to the phone screen—for example, to act on a permission request—then the app must display a message instructing the user to only look at their phone screen when it’s safe to do so.
	Sensitive Data	Browsers	SD-1: Before syncing data to the car, browsers that synchronize passwords or payments data must do the following steps: prompt the user to authenticate; notify the user on the car screen their data will be synchronized to the car.
			SD-2: Browsers must not save or allow access to passwords or payment information unless the user can block access to passwords using a profile lock.
	Driver Distraction	Video,Games,Browsers	DD-2: While driving, the app must not be launchable and the app’s UI must not be visible. The app’s audio must stop when the user starts driving and cannot be unpaused while driving.
			DD-3: The app must not be launchable or usable while driving and must not play any audio. Note: If your app plays audio, it must meet the DD-2 requirement.
	Voice Commands	Media,Navigation	VC-1: The app must support Google Assistant voice commands.
	Payments	Navigation,POI, IOT,Weather	PA-1: The app must have simple flows if purchases are enabled, using shortcuts such as recent or favorite purchases. The app must not allow any of the following: Setup of payment methods; Multiple items to be selected for purchase; Commitment to recurring payments, such as subscriptions.
	Media Autoplay	Media	MA-1: Media app must not autoplay on startup or without user initiated action to select the app or app media.
	Messaging Functionality	Messaging	MF-3: The user can reply to a message.
			MF-4: The app must use short-form messaging app design patterns. Traditional long-form messaging apps, such as apps for email, are not permitted.
	Navigation Functionality	Navigation	NF-5: The app must not provide turn-by-turn notifications, voice guidance, or cluster information when another navigation app is providing turn-by-turn instructions.
	Weather Functionality	Weather	NF-6: The app must handle navigation requests from other Android Auto apps.
Visual Effects	Contrast	Media,Messaging,POI, Navigation,IOT,Weather	VD-1: The app must provide icons and colors that meet Android Auto’s contrast requirements.
	User Experience	Video,Games,Browsers	UX-1: Touch targets are at least 64dp.
			UX-2: Touch targets are at least 24dp apart from each other and 24dp away from screen edges.
			UX-3: Font sizes should be at least 24sp.